**Code No: 114CX** **R13**

**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD**
**B.Tech II Year II Semester Examinations, December - 2017**
**JAVA PROGRAMMING**
**(Common to CSE, IT)**

**Time: 3 Hours** **Max. Marks: 75**

**Note:** This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b, c as sub questions.

**PART- A**

**(25 Marks)**

1.a) Define data abstraction. [2]
b) What is the size of char data type? Why does it differ from C language? [3]
c) What is the use of anonymous inner class? [2]
d) What is a package? How to define it and access it? [3]
e) Differentiate between error and exception. [2]
f) How to assign priorities to threads? [3]
g) List the functions of Stack class. [2]
h) What is the need of JDBC type 3, type 4 drivers? [3]
i) What are the sources for item event? [2]
j) Give the hierarchy for swing components. [3]

**PART-B**

**(50 Marks)**

2.a) What feature of Java makes it platform independent and portable?
b) Is Java a robust language? Justify your answer. [5+5]

**OR**

3.a) Differentiate between a class and object.
b) Demonstrate constructor overloading concept. [5+5]

4. What is inheritance? Explain different forms of inheritance with suitable program segments and real world example classes. [10]

**OR**

5.a) Differentiate between interface and abstract class.
b) What is meant by dynamic method dispatch? [5+5]

6.a) Write a program to illustrate the use of multiple catch blocks for a try block.
b) What are the uses of ‘throw’ and ‘throws’ clauses for exception handling? [5+5]

**OR**

7.a) What is the difference between a thread and a process?
b) How to achieve synchronization among threads? Write suitable code. [5+5]

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8.a) What is a vector? How does it differ from array, list?
b) Write a program to count number of words in a given sentence. [5+5]

**OR**

9.a) Write a program to copy the contents of file1 to file 2. Read the names of files as command line arguments.
b) Write about driver manager class for database connectivity. [5+5]

10.a) Why swing components are preferred over AWT components?
b) What is an adapter class? What is their role in event handling? [5+5]

**OR**

11.a) Explain the life cycle of an applet.
b) What are the various layout managers used in Java? [5+5]

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